Music Genre Classification with PCA

Reproducibility & Traceability Report

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Prepared for: Data Science Replication and Validation

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1. Data Provenance

Dataset: music_dataset_mod.csv
Source: Internal (no external APIs used).
Description: Contains numerical features describing musical properties with target variable **Genre**.

Initial Data Checks:
- Verified dataset shape, column types, and missing values.
- Missing values in **Genre** dropped before training; predicted post-modeling.

Target Variable: 'Genre' (encoded using LabelEncoder).

2. Analysis Workflow

1. **Data Cleaning:** Removed missing Genre entries; standardized numeric features with StandardScaler.
2. **Feature Engineering:** Verified multicollinearity via correlation matrix.
3. **PCA:** Reduced dimensionality; retained components explaining ≥80% variance.
4. **Model Training:** Logistic Regression (max_iter=10000) using both raw and PCA-transformed data.
5. **Evaluation:** Accuracy, Precision, Recall, and F1-score computed.
6. **Unknown Genre Prediction:** Applied trained model to fill missing Genre values.

3. Reproducibility Requirements

Environment:
Python 3.10+
Required Libraries: pandas, numpy, scipy, scikit-learn, matplotlib, seaborn

Randomness Control: random_state=42 ensures deterministic results.
Inputs: music_dataset_mod.csv
Outputs: Model performance reports and filled dataset (df_filled).

4. Threats to Validity

Threat Type	Description	Mitigation
Internal Validity	Preprocessing errors may bias outcomes.	Consistent use of fitted encoders/scalers; random seed fix
External Validity	Dataset may not represent global genre diversity.	Validate on multiple datasets.
Construct Validity	Features may omit cultural or lyrical elements.	Add richer audio/metadata features.
Statistical Validity	PCA may oversimplify data.	Compare PCA vs non-PCA performance.
Model Generalizability	Logistic Regression assumes linearity.	Explore nonlinear models (SVM, Random Forest).

5. Replication Guide (Summary)

Steps:
1. Install dependencies:

```
pip install pandas numpy scipy scikit-learn matplotlib seaborn
```

2. Place files: *music_dataset_mod.csv* and *Music Genre Classification with PCA - Project.py* in the same directory.
3. Run:

```
python "Music Genre Classification with PCA - Project.py"
```

4. Check console output for performance metrics and updated dataset.